

Sumit Roy

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OBJECTIVE

AI/ML engineering undergraduate with hands-on experience building LLM-powered systems — multi-agent orchestration, semantic search, and prompt engineering workflows. Comfortable working across the stack (Python, FastAPI, React) to take AI prototypes from idea to working product. Looking to apply this to real learner-facing AI systems at scale.

EDUCATION

B. Tech in Computer Science Engineering — CGPA: 7.99/10

Pillai College of Engineering, University of Mumbai

2023 – 2027

Panvel, Maharashtra

TECHNICAL SKILLS

- **AI/LLM Engineering:** LangGraph (Multi-Agent Systems), Gemini API, OpenAI API, Anthropic Claude API, Model Context Protocol (MCP), Function Calling, Sentence Transformers, Semantic Search, HyDE, Prompt Engineering, LLM-as-a-Judge Evaluation
- **Computer Vision:** OpenCV, Tesseract OCR
- **Programming Languages:** Python, JavaScript, TypeScript, C++
- **Frameworks & Libraries:** FastAPI, React.js, Next.js, Node.js
- **Databases:** PostgreSQL, MySQL, SQLite, MongoDB
- **Tools & Technologies:** Git, GitHub, Docker, REST APIs, Firebase, Alembic
- **Core Concepts:** Data Structures & Algorithms, OOP, DBMS, Operating Systems
- **Certifications:** Google Skills — Diamond League (10,600+ pts): Transformer Models & BERT, AI Agent Fundamentals, NLP on Google Cloud, DeepMind LLM Training/Fine-Tuning Series | AWS — Fundamentals of Machine Learning and Artificial Intelligence

EXPERIENCE

Computer Vision Intern (Co-developed)

Pillai College of Engineering

Dec 2024 – Jan 2025

Panvel, Maharashtra

- Engineered an end-to-end computer vision pipeline for automated manufacturing drawing analysis, jointly designing the preprocessing, detection, and classification stages with a project partner.
- Implemented an entity detection system using OpenCV and Tesseract OCR — applying grayscale conversion, Otsu's and adaptive thresholding, and morphological operations to improve text extraction accuracy.
- Categorized detected dimensions (diameter, radius, linear) and applied edge detection to identify structural lines, visualizing results with labelled bounding boxes.
- Identified and documented key limitations (e.g., arrow-line misclassification, duplicate-value disambiguation) to guide future improvements — hands-on experience evaluating model output quality.

Tech Stack: Python, OpenCV, Tesseract OCR

PROJECTS

Let Him Cook! — Agentic AI Cooking Simulator

LangGraph, FastAPI, Next.js, MySQL

(Solo) June 2026

- Architected a multi-agent orchestration system using LangGraph, coordinating five specialized agents (Planner, Inventory, Shopping, Cook, Judge) through a shared workflow state.
- Benchmarked multiple LLM providers (Anthropic Claude, Gemini, Groq) for agent reasoning, informally benchmarking output quality across providers to guide implementation choices.
- Developed a persistent backend (FastAPI, SQLAlchemy, MySQL) tracking wallets, inventory, transactions, and full workflow history, with a JWT-authenticated Next.js frontend.

Library Management System — AI-Powered Semantic Search

FastAPI, Sentence Transformers, Gemini 2.5, React

(Team Lead, 12 members) April 2026

- Led a 12-member team to deliver a full-stack library platform; personally, owned the AI/NLP backend.
- Validated the system using real records of 300+ students across multiple departments and a 350+ book catalog, validating functionality at realistic scale.
- Engineered a semantic search engine using Sentence Transformer embeddings, improving book discovery over traditional keyword search.
- Applied HyDE (Hypothetical Document Embeddings) to improve retrieval accuracy on ambiguous queries.
- Incorporated Google Gemini 2.5 for intent extraction and result summarization; project achieved full evaluation marks.

Knowledge Navigator Agent

Python, Gemini API, FastAPI, React

(Solo) Dec 2025

- Developed a ReAct-style (Reasoning + Action) LLM agent using the Gemini API to plan personalized course learning paths from real-time data.
- Connected Model Context Protocol (MCP) and custom function-calling schemas to connect the agent with live course retrieval and price comparison tools, avoiding static/hallucinated responses.
- Implemented session-based state management to retain user constraints (budget, skill level) across multi-turn conversations.
- Evaluated LLM-as-a-Judge and trace-based evaluation techniques to measure groundedness, tool-use accuracy, and decision quality.

ACTIVITIES & ACHIEVEMENTS

- Participated in Smart India Hackathon, Mumbai Hacks Hackathon, and attended E-Summit'25 at IIT Bombay
- Completed certifications: MATLAB Course, CSI Coding Essentials, Modern Sensors Technology & Industrial Applications